Class 9

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Background

We are going to learn more about principal component analysis (PCA) using a fun data set.

Data Import

```r
candy_file <- "candy-data.csv"

candy = read.csv("candy-data.txt", row.names=1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
```

```
Air Heads             0       1        0               0       0                 0
Almond Joy            1       0        0               1       0                 0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0    1        0        0.732        0.860   66.97173
3 Musketeers    0    1        0        0.604        0.511   67.60294
One dime        0    0        0        0.011        0.116   32.26109
One quarter     0    0        0        0.011        0.511   46.11650
Air Heads       0    0        0        0.906        0.511   52.34146
Almond Joy      0    1        0        0.465        0.767   50.34755
```

```r
#Imports our data
```

Q1. There are 85 different types of candy in the dataset

```r
#each row is a type of candy
nrow(candy)
```

```
[1] 85
```

Q2. There are 38 fruity candies in the data set

```r
#Counting the sum of fruity candy column should give us the amount
sum(candy$fruity)
```

```
[1] 38
```

What is the favorite candy

Q3. My favorite candy is Swedish Fish and the winpercent is 54.86111

Q4. The winpercent of Kit Kat is 76.7686

Q5. The winpercent of Tootsie Roll Snack Bars is 49.6535

```r
candy["Swedish Fish", ]$winpercent
```

```
[1] 54.86111
```

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

```r
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

```r
#Find the percent for each of the candy by looking at the candy data then for the row in the
```

Q6. The winpercent column looks like its on a different scale as every other variable is in between 0-1 while winpercent has numbers that are greater than 10

Q7. A zero most likely represents 'no' or that there is not any chocolate in the candy while a 1 represents a 'yes' where there is chocolate in the candy

```r
#install.packages("skimr")
library("skimr")
#lets skim the whole data det
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

```r
#lets look at just one varriable
skim(candy$chocolate)
```

Table 3: Data summary

Name	candy$chocolate
Number of rows	85
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

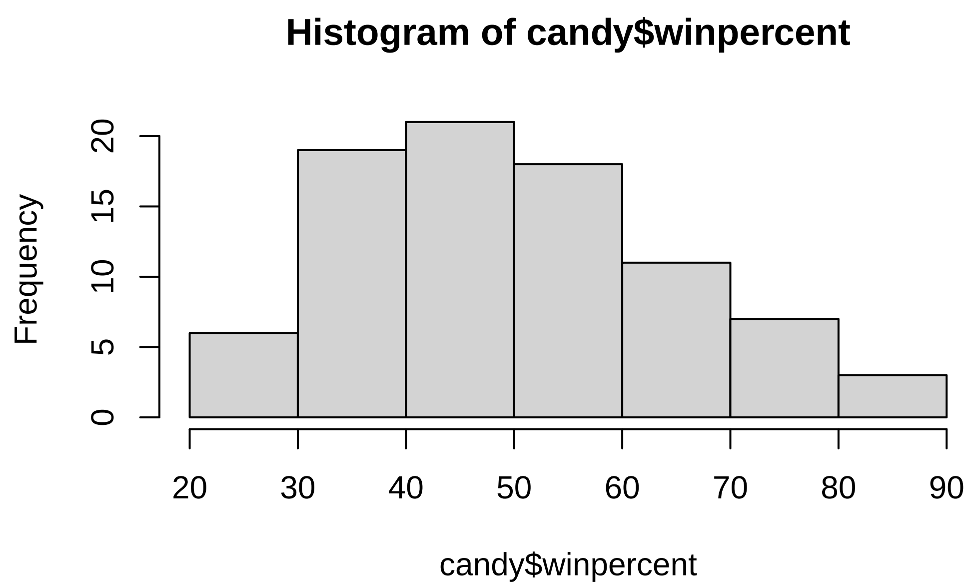
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	0	1	0.44	0.5	0	0	0	1	1	

Exploratory Analysis

Lets look at the histograms

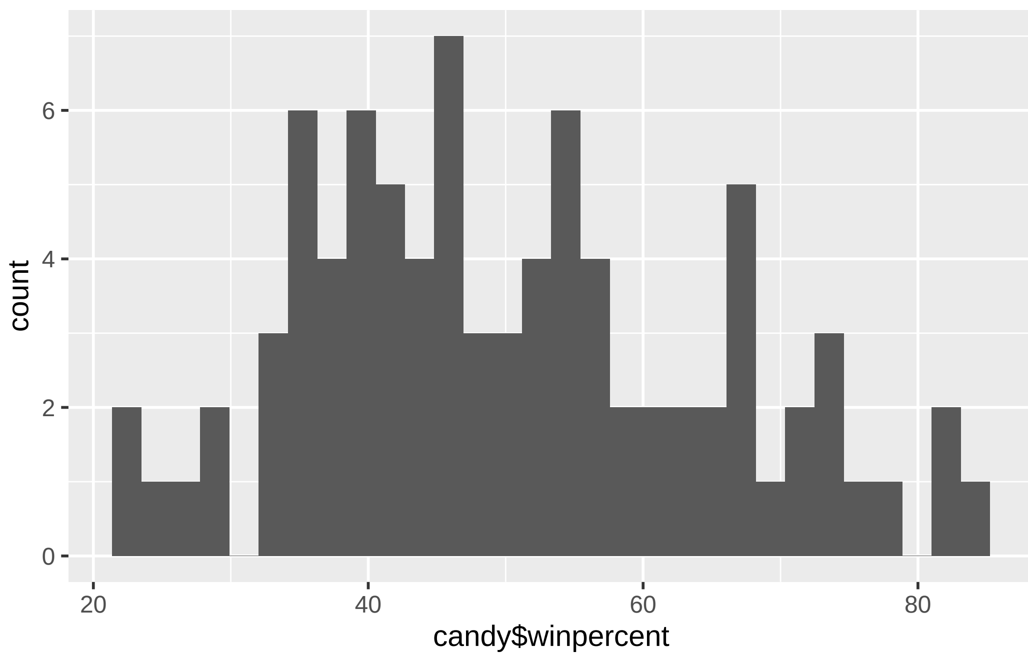
Q8.

```r
hist(candy$winpercent)
```




```r
library(ggplot2)
ggplot(candy)+
  aes(candy$winpercent)+
  geom_histogram()
```

```
Warning: Use of `candy$winpercent` is discouraged.
i Use `winpercent` instead.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
#Makes a plot
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

Q.9 The distribution isn't symmetrical, it is slightly skewed to the left

Q.10 The center of distribution is slightly above 50% at 50.31676 with the mean, but looking at the median which would be better for skewed data like this, it is at 47.83 or bellow %50

Q.11 The average chocolate candy is higher ranked at 60.92153 than fruity candy at 44.11974

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

```
#we can use as.logical() since these are T/F values
```

Q12. The p-value is 2.871e-08 which is much smaller than 0.05 indicating that it is significant

```
t.test((candy$winpercent[as.logical(candy$chocolate)]),candy$winpercent[as.logical(candy$fru
```

```
	Welch Two Sample t-test

data:  (candy$winpercent[as.logical(candy$chocolate)]) and candy$winpercent[as.logical(candy$
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

```
#uses a t-test function
```

Candy Rankings

Q.13 The least favorite candies are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters

```
head(candy[order(candy$winpercent),], 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
```

```
                    winpercent
Nik L Nip             22.44534
Boston Baked Beans    23.41782
Chiclets              24.52499
Super Bubble          27.30386
Jawbusters            28.12744
```

```
#uses the head function to find the candies, the (,5) selects 5
```

Q14. The top 5 are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers

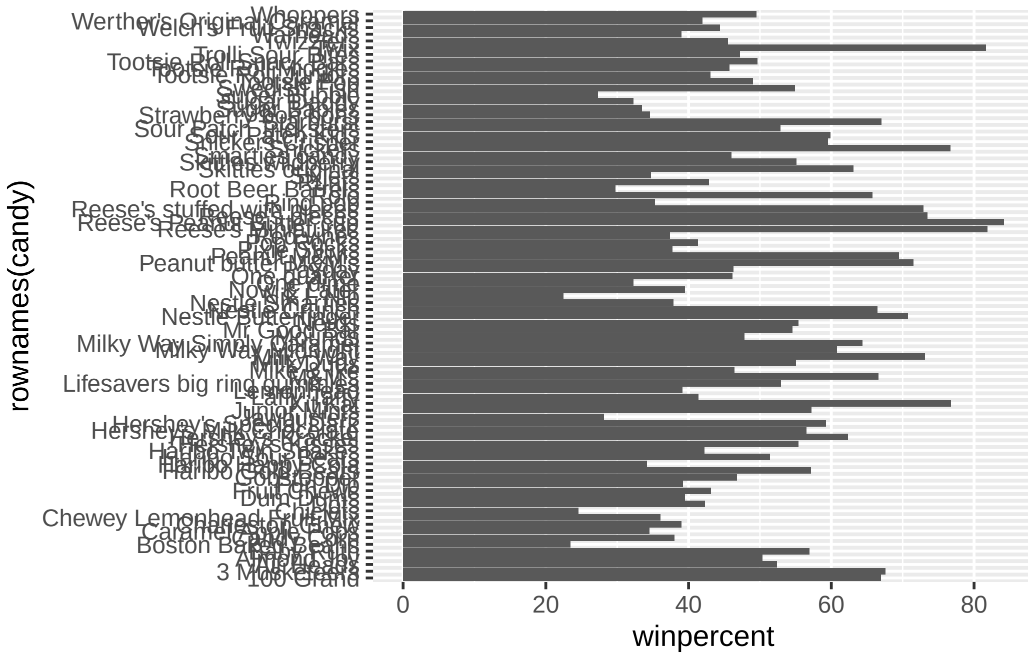
```
head(candy[order(candy$winpercent, decreasing = T),], 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

```
#For the least popular
```

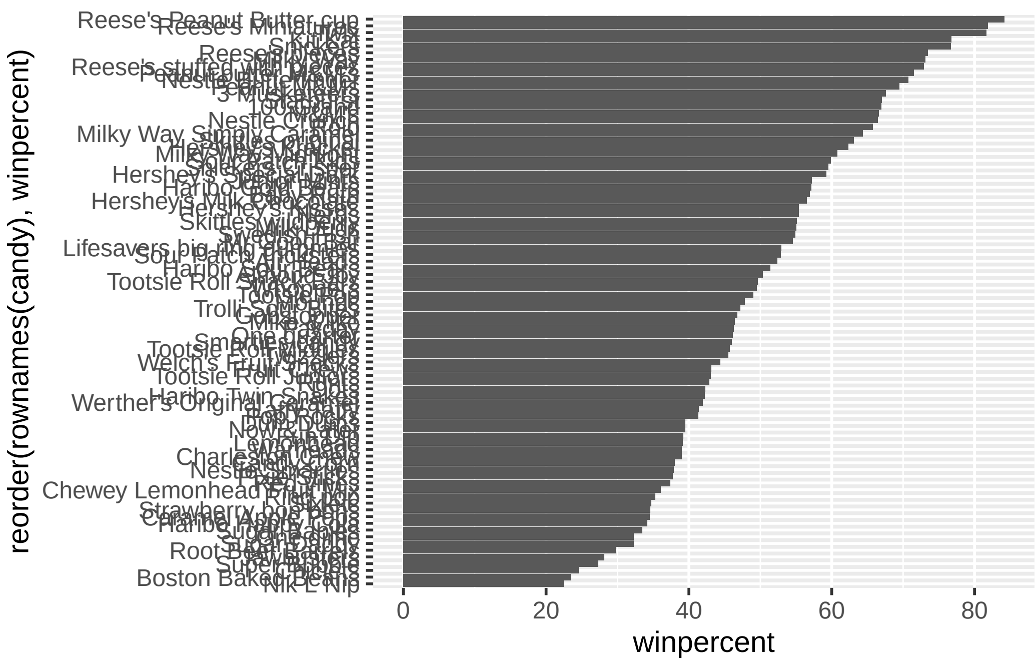
Q15.

```
library(ggplot2)
#Makes a plot
ggplot(candy) +
  aes(winpercent, rownames(candy)) + geom_col()
```



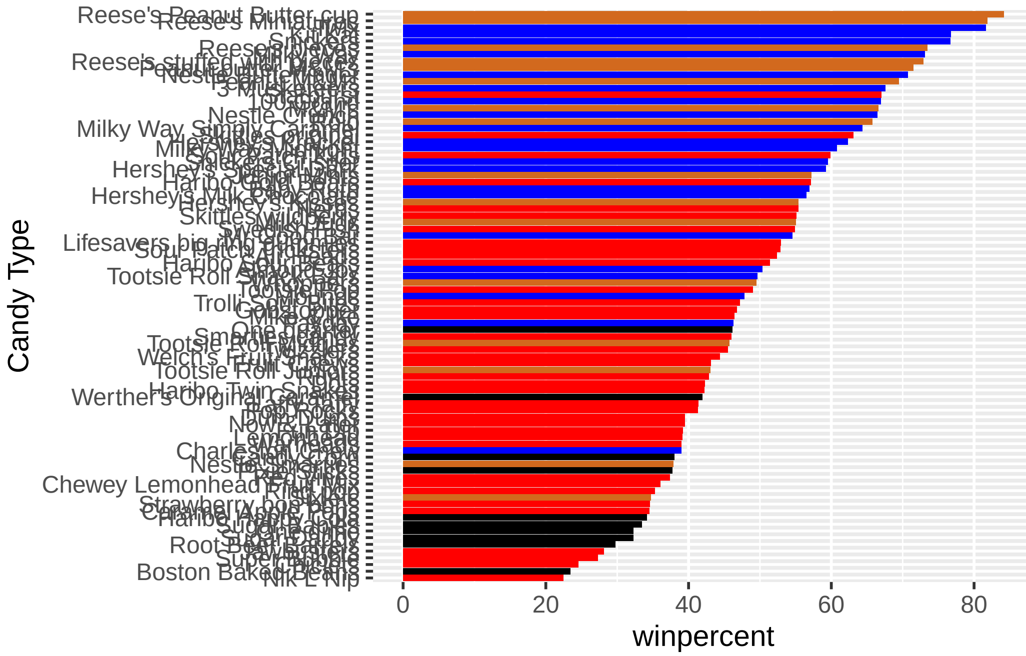
Q16.

```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) + geom_col()
```



Adding Color

```
#set all colors
my_cols=rep("black", nrow(candy))
#chocolate
my_cols[as.logical(candy$chocolate)] = "chocolate"
#bars
my_cols[as.logical(candy$bar)] = "blue"
#fruit
my_cols[as.logical(candy$fruity)] = "red"
#Adds colors to each type of candy for the following plot
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) + geom_col(fill=my_cols) +
  ylab("Candy Type")
```



Q17. The worst ranked chocolate is Sixlets

Q18. The best fruity candy is Starburst

Pricepercent

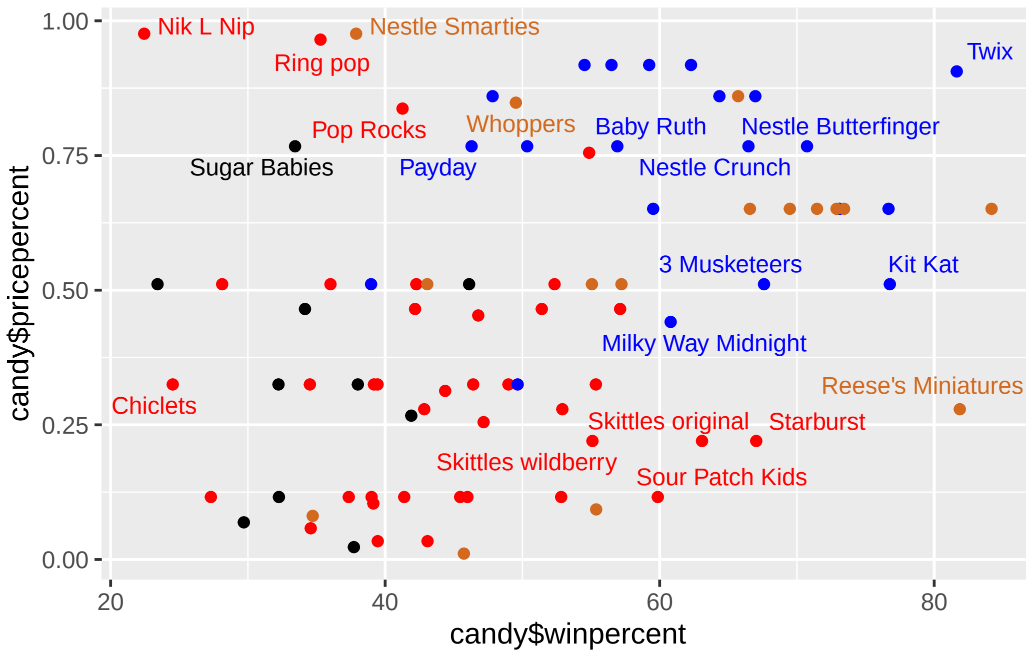
```r
#install.packages("ggrepel")
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(candy$winpercent, candy$pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
Warning: Use of `candy$winpercent` is discouraged.
i Use `winpercent` instead.
```

```
Warning: Use of `candy$pricepercent` is discouraged.
i Use `pricepercent` instead.
```

```
Warning: Use of `candy$winpercent` is discouraged.
i Use `winpercent` instead.
```

```
Warning: Use of `candy$pricepercent` is discouraged.
i Use `pricepercent` instead.
```



Q19. It would be Twix

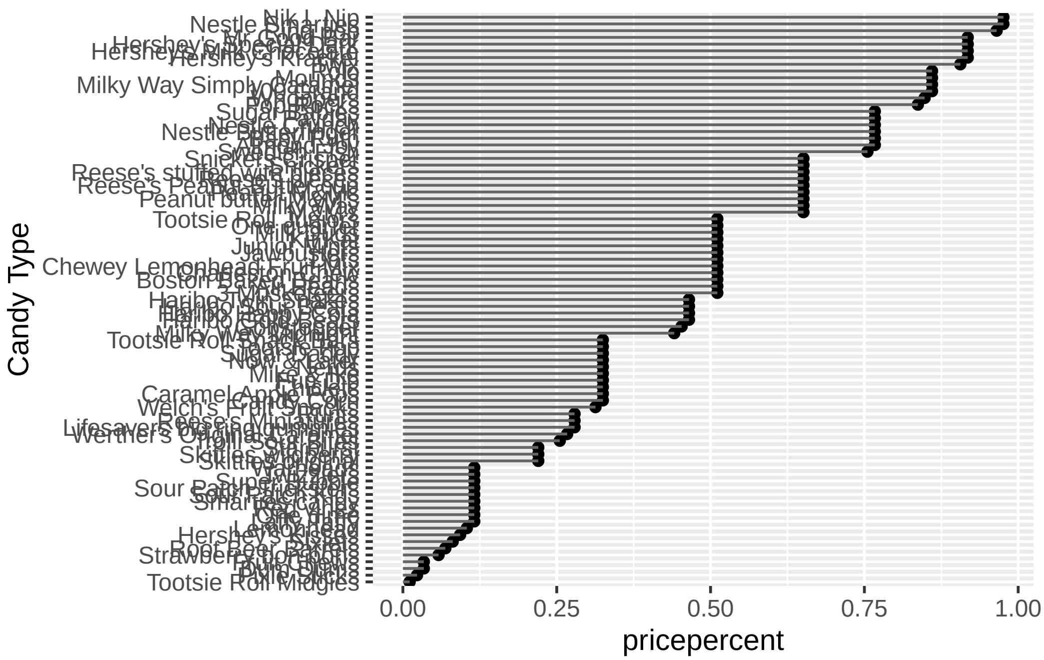
Q20. The most expensive candies are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, Hershey's Milk Chocolate. With Nik L Nip being the least popular

```r
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

Q21.

```r
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) + geom_point() +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent), xend = 0), col="gray40")+
  ylab("Candy Type")
```

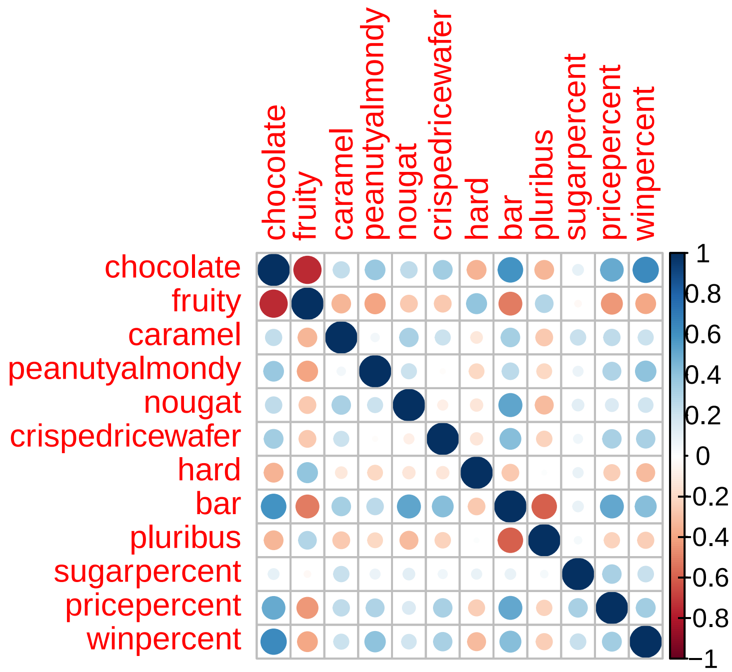


Correlation Structure

```r
#install.packages("corrplot")
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
#finds the correlation between each pair of variables
corrplot(cij)
```



Q22. The strongest anti-correlation is fruity and chocolate

Q23.

Since chocolate and fruit are so anti-correlated, I predict that they will have the largest contribution to PC1

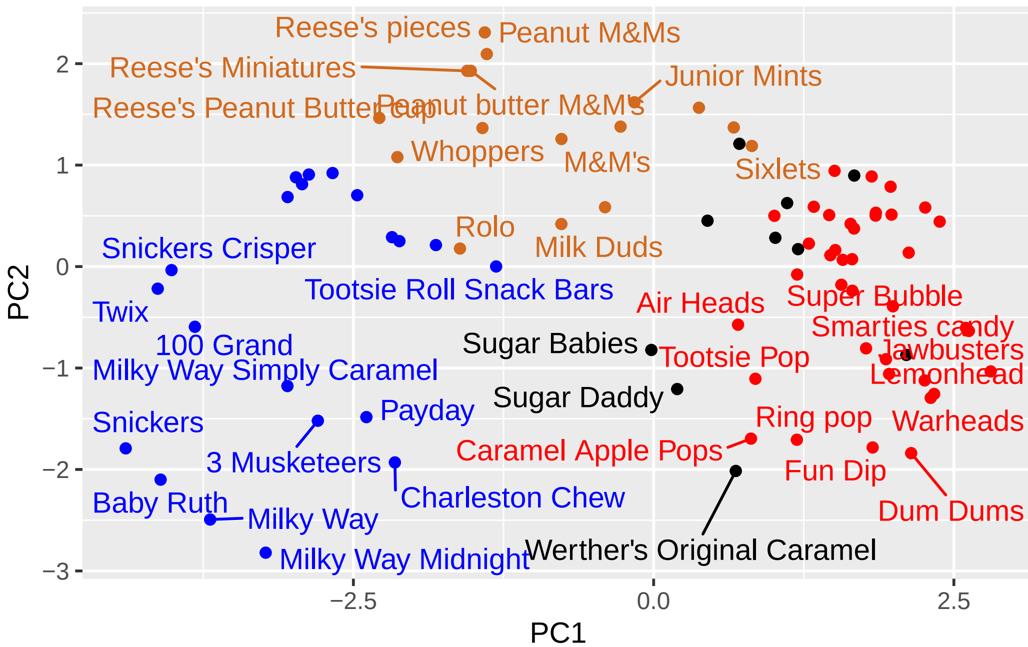
PCA

```
pca<- prcomp(candy, scale=T)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

```
#makes pca of the data, using the scale= T to ensure that the size of winpercent doesn't ove
```

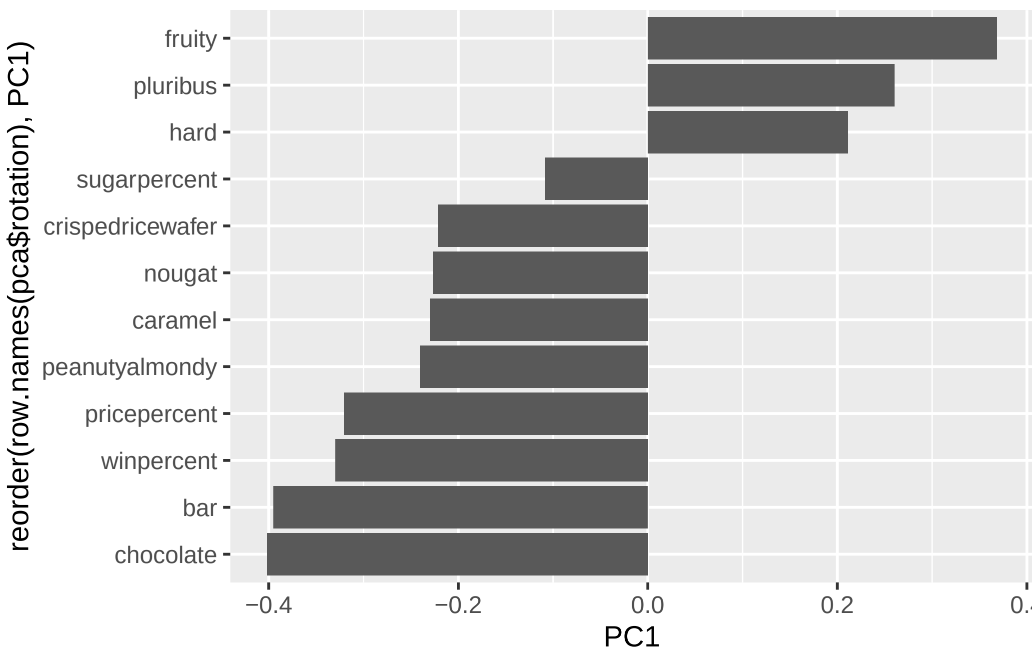
```
ggplot(pca$x)+
  aes(PC1,PC2, label = row.names(pca$x))+ geom_point(col=my_cols)+
  geom_text_repel(col=my_cols)
```




```
#graphs the PCA
```

Second major results figure from PCA

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. The variables of fruity followed by pluribus and hard were strongly picked up by PC1 in the postive direction. This makes sense to me, especially as we saw in the correlation plot how fruity had the strongest minus correlation with chocolate and a lot of negative correlation with other variables

Q25. I think I would make a chocolate and bar as that is clearly the winner. Firstly, looking at the graph of candy type against winpercent, we see the top of the list occupied by chocolates and bars as evident of the brown and blue near the top of chart. Next, in our correlation chart, we see the highest levels of correlation between winpercent is with chocolate, with bar also being blue but at a lighter shade. Finally, looking at PCA, such as loading, we see chocolate, bar, and winpercent close to one another indicating their association. With all this data in mind, chocolate and bar is a winning candy.